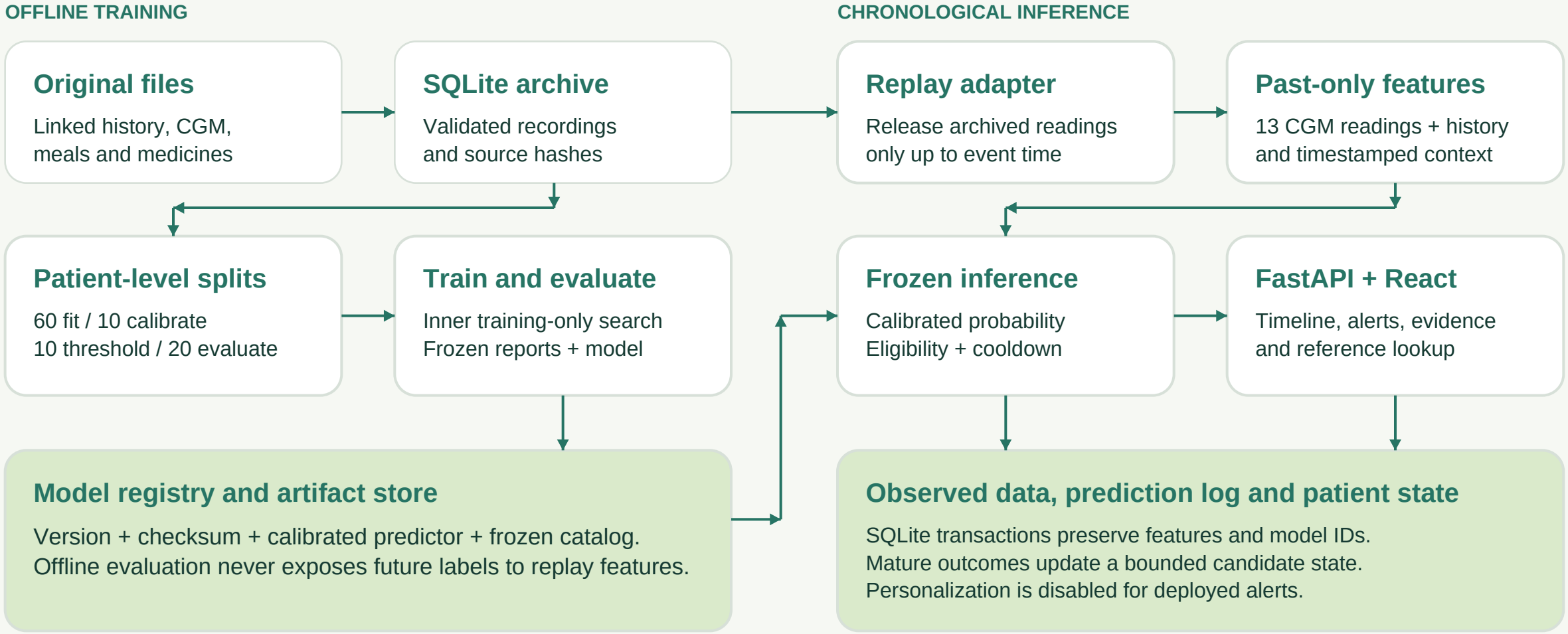
Architecture: one app, two execution paths



One local FastAPI worker serves the API and built UI. Training runs as a separate CLI job. No live wearable connection.

Data relationships and chronology safeguards

Patient -> Recording

One person can have several recordings.
All visits stay in one split within a run.

Recording -> Source data

SourceReading + ContextEvent
CGM and timestamped meals/medicines.

ModelVersion -> Replay

Each replay pins one model.
Later activation does not rewrite history.

Replay -> ObservedReading

Only released readings enter features.
Insufficient or broken history means no new prediction.

Replay -> Prediction -> Alert

Save exact features, probabilities, model version and state snapshot.
Alerts can be acknowledged.

Mature outcomes -> State

Resolve complete outcomes at +75 min.
Spaced updates; reset at gaps/visits.
Global model weights stay frozen.

Additional stores: NutritionCatalog / NutritionReference preserve lookup provenance; AuditEvent records important operations.
Patient state is stored with the replay and prediction snapshots.